

Multiple Choice (10 marks)

1. Substances whose Lewis structures must be drawn with an unpaired electron are called
 - (a) hydrogen bonds
 - (b) free radicals
 - (c) allotropes
 - (d) isotopes
 - (e) nonpolar molecules
2. Consider the Lewis structures for the following molecules: CO_2 , CO_3^{2-} , NO_2^- , and NO_3^- . Which molecule would have the shortest bonds?
 - (a) O_2
 - (b) NO_3
 - (c) NO_2^-
 - (d) CO_3^{2-}
 - (e) NO_2
3. What is the NMR frequency of ^{31}P in a 20.0 T magnetic field?
 - (a) 310.2 MHz
 - (b) 500.5 MHz
 - (c) 275.0 MHz
 - (d) 400.0 MHz
 - (e) 120.2 MHz
4. Monatomic ions of the representative elements are often
 - (a) very electronegative
 - (b) typically metallic
 - (c) highly colored
 - (d) mostly gases at room temperature
 - (e) often very reactive with water
5. Using fundamental trends in electronegativity and bond strength, which of the following should be the strongest acid?
 - (a) H_2Se
 - (b) H_2CO_3

- (c) H_2S
- (d) HBr
- (e) H_2O

True / False (10 marks)

Answer True or False

6. True or False: The energy of yellow light from a sodium lamp is approximately 1.5 eV and corresponds to about 35.2 Kcal/mole.
7. True or False: The swimmer's metabolic rate increases upon exiting the cold water, contributing to the sensation of warmth.
8. True or False: The reaction between MgBr_2 and Na_2CO_3 produces a net ionic equation with distinct ions remaining in solution.
9. True or False: The angular-momentum quantum number l for a t orbital is 5.
10. True or False: For the reversible reaction $\text{H}_2 + \text{X}_2 \rightleftharpoons 2 \text{HX}$, the thermodynamic equilibrium constant K can be expressed as (k_a / k_b) .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that determine the limiting high-temperature molar heat capacity at constant volume (C_V) for a gas-phase diatomic molecule and explain why it equals $4.5 R$.
12. Discuss the relationship between the potential energy and de Broglie wavelength of a proton initially at rest. Calculate the potential through which it must fall to achieve a wavelength of $1.0 \times 10^{-10} \text{ m}$.

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